READ ME FILE – Crime Data Pipeline

Project Overview

This project was developed as part of a Python Data Engineering assignment focused on preparing UK police crime data for downstream Business Intelligence reporting.

The project simulates the role of a Data Engineer working within a Police Force Analytics Unit.

The objective of the pipeline is to:

- ingest large operational crime datasets
- clean and validate raw data
- enrich crime data with contextual datasets
- aggregate the data to a reporting-ready grain
- export a clean dataset for Power BI analysis

Datasets Used

1. UK Police Crime Data (Primary Dataset)

Source: https://data.police.uk/data/

Police forces selected:

- Metropolitan Police
- Merseyside Police
- West Midlands Police
- Northumbria Police
- (Would have also used Greater Manchester Police but the data was unavailable. "Greater Manchester Police: Currently no crime, outcome or stop and search data is available. The force is progressing towards a new IT system and data processes which will support provision of data to police.uk in the future.") https://data.police.uk/changelog/

Data includes:

- crime category
- reporting month
- geographic identifiers
- outcome information

Where to find:

- File called "Crime Data" in the sharepoint.
- https://harnham.sharepoint.com/sites/RockborneProgrammeTrainingMaterials/Training%20Materials/Forms/AllItems.aspx?id=%2Fsites%2FRockborneProgrammeTrainingMaterials%2FTraining%20Materials%2FCH%2019%20Submission%2F05%20Python%20for%20Data%20Analysis%2FProjects%2FCallumFisher%5F05PythonForDataAnalysis%2FCrime%20Data&viewid=2688a5c0%2D757d%2D4b6f%2D8161%2D248245deafc6&p=true

2. Football Match Dataset (Enrichment Dataset)

Source: https://www.football-data.co.uk/

Purpose: Football match activity was used to enrich the crime data and support analysis of whether football events influence crime levels.

Measures created:

- home_match_count
- derby_match_count

Where to find:

- File called "Football Data" in the sharepoint.
- https://harnham.sharepoint.com/sites/RockborneProgrammeTrainingMaterials/Training%20Materials/Forms/AllItems.aspx?id=%2Fsites%2FRockborneProgrammeTrainingMaterials%2FTraining%20Materials%2FCH%2019%20Submission%2F05%20Python%20for%20Data%20Analysis%2FProjects%2FCallumFisher%5F05PythonForDataAnalysis%2FFootball%20Data&viewid=2688a5c0%2D757d%2D4b6f%2D8161%2D248245deafc6&p=true

3. Population Data (Enrichment Dataset)

Source: Office for National Statistics (ONS)

Purpose: Population estimates were used to create normalised crime-rate metrics.

Measures created:

- population
- crime_rate_per_1000

Where to find:

- File called "Population Data" in the sharepoint
- https://harnham.sharepoint.com/sites/RockborneProgrammeTrainingMaterials/Training%20Materials/Forms/AllItems.aspx?id=%2Fsites%2FRockborneProgrammeTrainingMaterials%2FTraining%20Materials%2FCH%2019%20Submission%2F05%20Python%20for%20Data%20Analysis%2FProjects%2FCallumFisher%5F05PythonForDataAnalysis%2FPopulation%20Data&viewid=2688a5c0%2D757d%2D4b6f%2D8161%2D248245deafc6&p=true

Final Reporting Grain

The final reporting dataset is aggregated at:

Police Force × Month × Crime Type

Each row represents:

- one police force
- one reporting month
- one crime category

with additional football and population enrichment metrics.

Pipeline Stages

1. Ingestion Layer

The pipeline imports:

- monthly crime CSV files
- football season CSV files

- population datasets

Data is processed iteratively rather than loading all raw data at once.

2. Cleaning and Validation Layer

The pipeline performs:

- column standardisation
- date cleaning
- missing value handling
- duplicate validation
- row-count validation
- reporting-grain validation

Validation checks include:

- row counts before and after cleaning
- duplicate checks at reporting grain
- null checks on key fields

3. Feature Engineering and Transformation

The pipeline creates:

- year fields
- month fields
- monthly aggregation metrics
- football match indicators
- derby-match indicators
- population-normalised crime rates

4. Aggregation Layer

Crime data is aggregated to:

Police Force × Month × Crime Type

This produces BI-ready metrics including:

- crime_count
- home_match_count
- derby_match_count
- crime_rate_per_1000

5. Exporting Layer

The final dataset is exported as:

final_reporting_dataset.csv

This file is designed for downstream Power BI reporting.

(Final crime dataset exported as: crime_monthy_clean.csv)

(Final football dataset exported as: football_monthly.csv)

(Final population dataset exported as: population_clean.csv)

Main Functions

Crime Functions:

- clean_column_names()
- clean_dates()
- process_crime_file()
- process_crime_files()
- aggregate_crime_monthly()

Football Functions:

- clean_football_columns()
- clean_football_dates()
- process_football_file()
- process_football_files()

- aggregate_football_monthly()

Population Functions:

- clean_population_columns()
- reshape_population_data()
- aggregate_population_by_force()
- process_population_data()

Merge and KPI Functions:

- merge_crime_and_football()
- merge_population()
- add_crime_rate()

Validation Checks

The following validation checks were implemented:

- duplicate row validation
- missing value validation
- reporting-grain validation
- row count reconciliation
- join validation

Assumptions

Population Assumption:

Annual population estimates were used for all months within the corresponding year.

This approach is acceptable because population changes slowly relative to monthly crime trends.

2025 Population Estimates:

2024 ONS population estimates were reused for 2025 because complete 2025 subnational estimates were not yet available.

Football Assumptions:

Football enrichment only includes selected Premier League teams associated with the chosen police-force regions. This could be improved by adding more leagues and more police forces.

Used dropna() as the football data that does not have a home or away team, or a date will be completely useless and there is no way to combat this.

Intended BI Usage

The final dataset supports:

- trend analysis
- seasonal analysis
- police-force comparisons
- crime-type analysis
- crime-rate analysis
- Power BI dashboards
- operational reporting

GitHub :

https://github.com/Callum-Fisher/Python-Cleaning-Project

```
callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub
$ mkdir git-PythonCleaningProject

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub
$ cd git-PythonCleaningProject

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject
$ git init
Initialized empty Git repository in C:/Rockborne/GitHub/git-PythonCleaningProject/.git/

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        CallumFisher_PythonForDataAnalysis_CleaningProject.ipynb
        CallumFisher_PythonForDataAnalysis_WorkflowDiagram.pdf

nothing added to commit but untracked files present (use "git add" to track)

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (maste
r)
$ git add CallumFisher_PythonForDataAnalysis_CleaningProject.ipynb
warning: in the working copy of 'CallumFisher_PythonForDataAnalysis_CleaningProject.ipynb', LF will be replaced by CRLF the next time Git touches it

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git commit -m"Python Code for Cleaning Crime,Football,and Population Data"
[master (root-commit) d0794df] Python Code for Cleaning Crime,Football,and Population Data
 1 file changed, 5107 insertions(+)
 create mode 100644 CallumFisher_PythonForDataAnalysis_CleaningProject.ipynb

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git remote add origin https://github.com/Callum-Fisher/Python-Cleaning-Project.git

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git remote -v
origin  https://github.com/Callum-Fisher/Python-Cleaning-Project.git (fetch)
origin  https://github.com/Callum-Fisher/Python-Cleaning-Project.git (push)

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git push origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 18.67 KiB | 3.11 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Callum-Fisher/Python-Cleaning-Project.git
 * [new branch]      master -> master
```

```
callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        CallumFisher_PythonForDataAnalysis_WorkflowDiagram.pdf

nothing added to commit but untracked files present (use "git add" to track)

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git add CallumFisher_PythonForDataAnalysis_WorkflowDiagram.pdf
warning: in the working copy of 'CallumFisher_PythonForDataAnalysis_WorkflowDiagram.pdf', LF will be replaced by CRLF the next time Git touches it

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git commit -m"Workflow Diagram"
[master 80051f6] Workflow Diagram
 1 file changed, 79 insertions(+)
 create mode 100644 CallumFisher_PythonForDataAnalysis_WorkflowDiagram.pdf

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git remote add origin https://github.com/Callum-Fisher/Python-Cleaning-Project.git
error: remote origin already exists.

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.19 MiB | 1.09 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Callum-Fisher/Python-Cleaning-Project.git
   d0794df..80051f6  master -> master

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        CallumFisher_PythonForDataAnalysis_DataDictionary.pdf
        CallumFisher_PythonForDataAnalysis_READMEfile.pdf
        CallumFisher_PythonForDataAnalysis_StatementOfWork.pdf

nothing added to commit but untracked files present (use "git add" to track)

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git add CallumFisher_PythonForDataAnalysis_DataDictionary.pdf

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git commit -m"Data Dictionary"
[master e823ea7] Data Dictionary
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 CallumFisher_PythonForDataAnalysis_DataDictionary.pdf
```

```
callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 32.55 KiB | 10.85 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Callum-Fisher/Python-Cleaning-Project.git
   80051f6..e823ea7  master -> master

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git add CallumFisher_PythonForDataAnalysis_StatementOfWork.pdf

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git commit -m"Statement of Work"
[master f3bfbe5] Statement of Work
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 CallumFisher_PythonForDataAnalysis_StatementOfWork.pdf

callu@CallumsLaptop MINGW64 /c/Rockborne/GitHub/git-PythonCleaningProject (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 82.54 KiB | 11.79 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Callum-Fisher/Python-Cleaning-Project.git
   e823ea7..f3bfbe5  master -> master
```

Added, Committed and Pushed this README file after the screenshot.